

labsheet=04 PRIYANSHU SHARMA CU240250980 SECTION(B) (47) Mastery in Data Analytics and Visualizations and Career Advancement (Technical)

```
# Advanced Data Visualization using Matplotlib and Seaborn
# Dataset: Superstore-style Sales Dataset
# =====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set visualization style
sns.set_theme(style="whitegrid")
```

```
# 1. CREATE / LOAD DATASET
# =====

print("=" * 60)
print("EXPERIMENT NO. 4")
print("ADVANCED DATA VISUALIZATION")
print("=" * 60)

# Sample Superstore dataset
data = {
    "Region": [
        "East", "West", "Central", "South", "East",
        "West", "Central", "South", "East", "West",
        "Central", "South", "East", "West", "Central"
    ],
    "Category": [
        "Technology", "Furniture", "Office Supplies", "Technology",
        "Furniture", "Office Supplies", "Technology", "Furniture",
        "Office Supplies", "Technology", "Furniture", "Office Supplies",
        "Technology", "Furniture", "Office Supplies"
    ],
    "Sales": [
        2500, 1800, 1200, 3200, 2100,
        1500, 2800, 1700, 1100, 3500,
        2200, 1300, 2900, 1900, 1400
    ],
    "Profit": [
        500, 300, 180, 700, 350,
        250, 600, 280, 160, 800,
        400, 200, 650, 320, 220
    ],
    "Quantity": [
        10, 7, 5, 12, 8,
        6, 11, 7, 4, 14,
        9, 5, 10, 8, 6
    ],
    "Customer_Age": [
        25, 32, 28, 45, 36,
        29, 41, 38, 24, 50,
        34, 27, 43, 31, 26
    ]
}
```

```
df = pd.DataFrame(data)

print("\nFirst 5 rows:")
print(df.head())

print("\nDataset Information:")
print(df.info())

print("\nBasic Statistics:")
print(df.describe())
```

```
=====
EXPERIMENT NO. 4
ADVANCED DATA VISUALIZATION
=====
```

First 5 rows:

	Region	Category	Sales	Profit	Quantity	Customer_Age
0	East	Technology	2500	500	10	25
1	West	Furniture	1800	300	7	32
2	Central	Office Supplies	1200	180	5	28
3	South	Technology	3200	700	12	45
4	East	Furniture	2100	350	8	36

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 15 entries, 0 to 14
```

```
Data columns (total 6 columns):
```

#	Column	Non-Null Count	Dtype
0	Region	15 non-null	object
1	Category	15 non-null	object
2	Sales	15 non-null	int64
3	Profit	15 non-null	int64
4	Quantity	15 non-null	int64
5	Customer_Age	15 non-null	int64

```
dtypes: int64(4), object(2)
```

```
memory usage: 852.0+ bytes
```

```
None
```

Basic Statistics:

	Sales	Profit	Quantity	Customer_Age
count	15.000000	15.000000	15.000000	15.000000
mean	2073.333333	394.000000	8.133333	33.933333
std	758.256145	206.356142	2.850230	7.995237
min	1100.000000	160.000000	4.000000	24.000000
25%	1450.000000	235.000000	6.000000	27.500000
50%	1900.000000	320.000000	8.000000	32.000000
75%	2650.000000	550.000000	10.000000	39.500000
max	3500.000000	800.000000	14.000000	50.000000

```
# 2. BAR CHART
```

```
# Compare sales across regions
```

```
# =====
```

```
print("\n" + "=" * 60)
```

```
print("2. BAR CHART - SALES BY REGION")
```

```
print("=" * 60)
```

```
region_sales = df.groupby("Region")["Sales"].sum()
```

```
print("\nSales by Region:")
```

```
print(region_sales)
```

```
plt.figure(figsize=(8, 5))
```

```
plt.bar(
    region_sales.index,
    region_sales.values
```

```
)

plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.grid(axis="y", alpha=0.3)

plt.tight_layout()
plt.show()
```

```
# 3. LINE CHART
# Analyze sales trend
# =====

print("\n" + "=" * 60)
print("3. LINE CHART - SALES TREND")
print("=" * 60)

df["Month"] = range(1, len(df) + 1)

plt.figure(figsize=(9, 5))

plt.plot(
    df["Month"],
    df["Sales"],
    marker="o",
    linewidth=2,
    label="Sales"
)

plt.title("Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.legend()

plt.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```

```
# 4. HISTOGRAM
# Distribution of sales
# =====

print("\n" + "=" * 60)
print("4. HISTOGRAM - SALES DISTRIBUTION")
print("=" * 60)

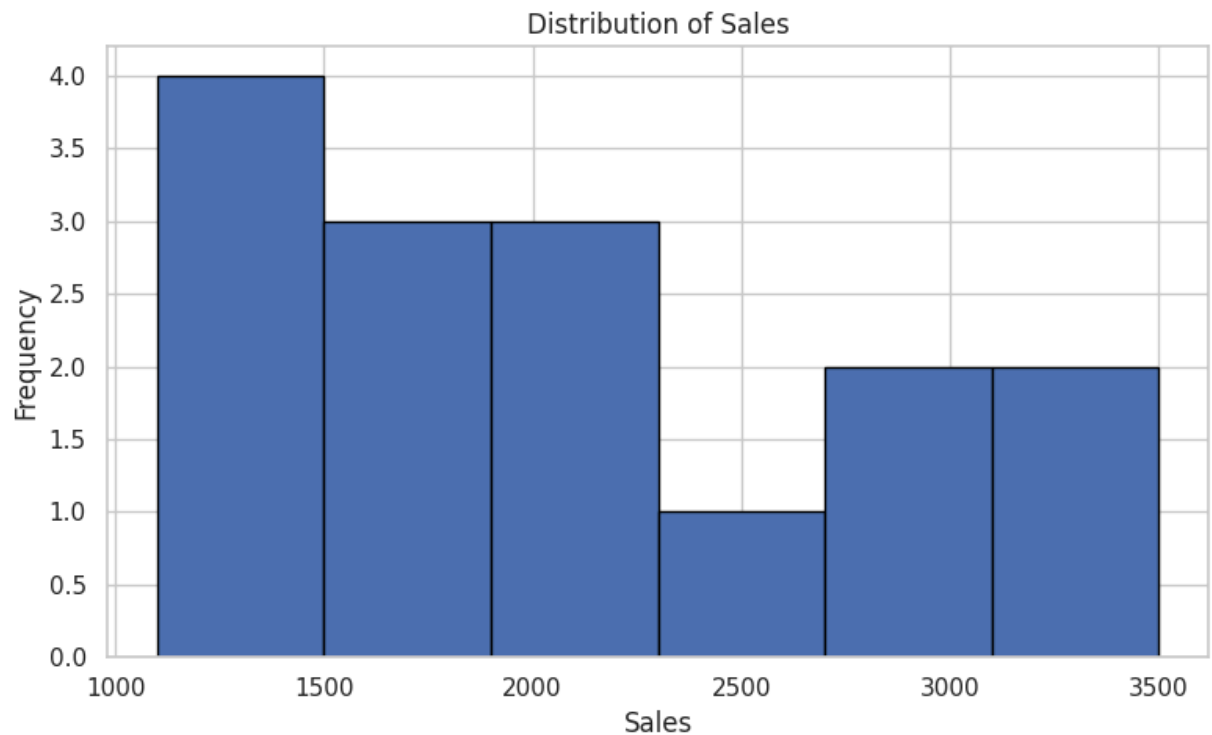
plt.figure(figsize=(8, 5))

plt.hist(
    df["Sales"],
    bins=6,
    edgecolor="black"
)

plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

```
=====
4. HISTOGRAM - SALES DISTRIBUTION
=====
```



```
# 5. BOX PLOT
# Identify outliers and distribution
# =====

print("\n" + "=" * 60)
print("5. BOX PLOT - SALES BY REGION")
print("=" * 60)

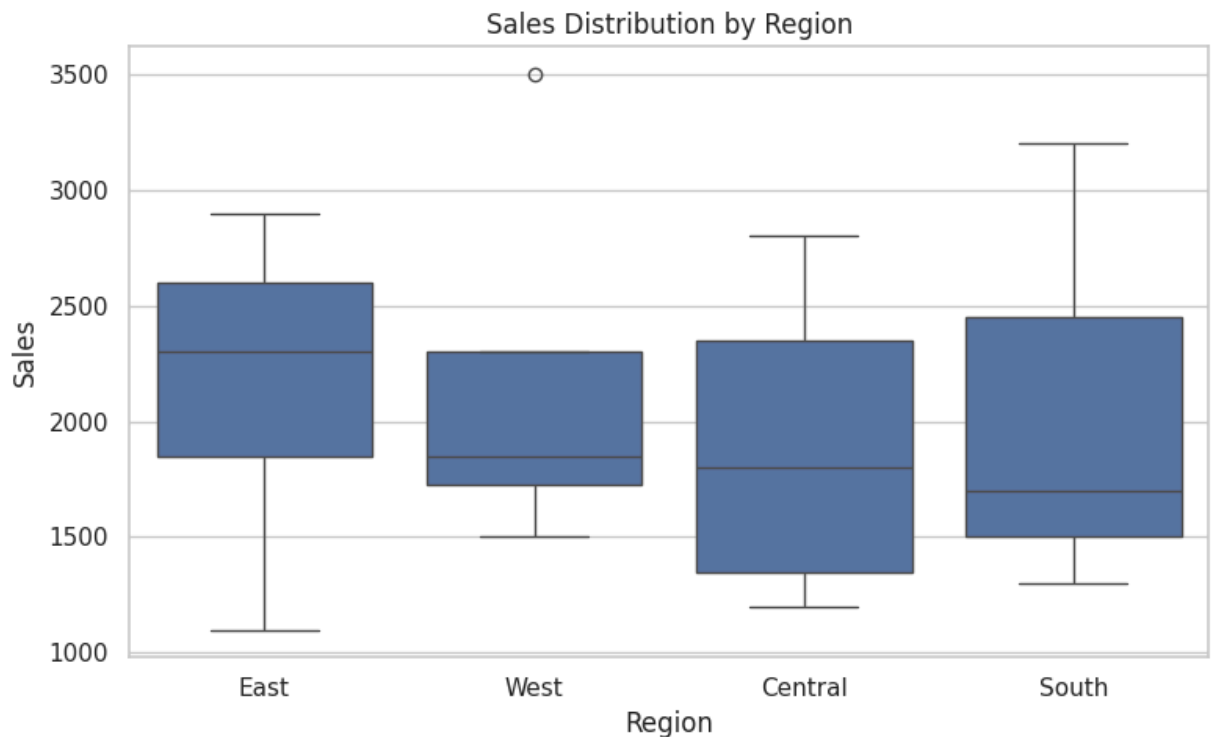
plt.figure(figsize=(8, 5))

sns.boxplot(
    data=df,
    x="Region",
    y="Sales"
)

plt.title("Sales Distribution by Region")
plt.xlabel("Region")
plt.ylabel("Sales")

plt.tight_layout()
plt.show()
```

```
=====
5. BOX PLOT - SALES BY REGION
=====
```



```
# 6. SCATTER PLOT
# Relationship between Sales and Profit
# =====

print("\n" + "=" * 60)
print("6. SCATTER PLOT - SALES VS PROFIT")
print("=" * 60)

plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit",
    hue="Category",
    s=100
)

plt.title("Relationship Between Sales and Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.legend(title="Category")

plt.tight_layout()
plt.show()
```

=====

6. SCATTER PLOT - SALES VS PROFIT

=====



7. CORRELATION HEATMAP

=====

```
print("\n" + "=" * 60)
print("7. CORRELATION HEATMAP")
print("=" * 60)

numeric_data = df[
    ["Sales", "Profit", "Quantity", "Customer_Age"]
]

correlation = numeric_data.corr()

print("\nCorrelation Matrix:")
print(correlation)

plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5
)

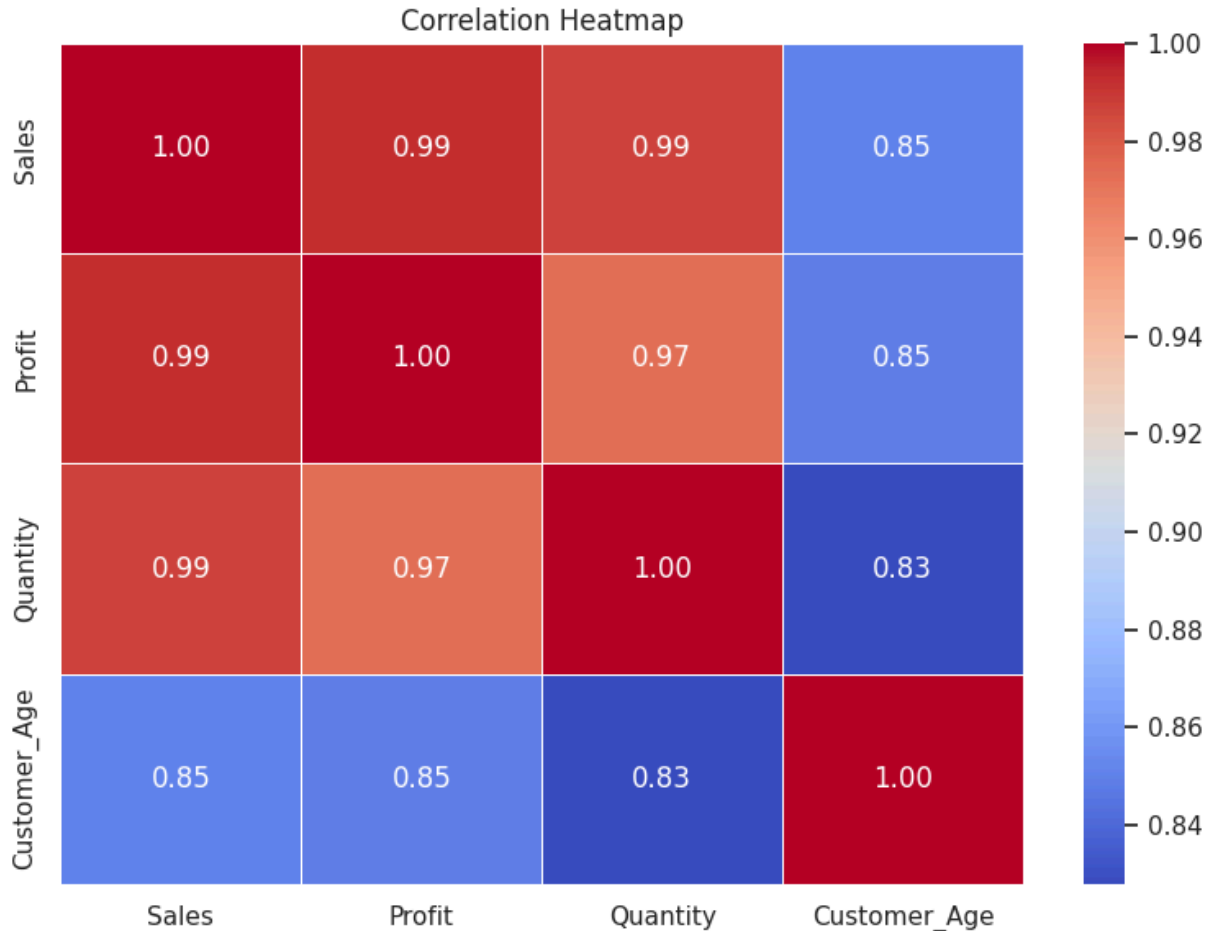
plt.title("Correlation Heatmap")

plt.tight_layout()
plt.show()
```

```
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7. CORRELATION HEATMAP
=====
```

Correlation Matrix:

	Sales	Profit	Quantity	Customer_Age
Sales	1.000000	0.993156	0.986663	0.850357
Profit	0.993156	1.000000	0.973006	0.849160
Quantity	0.986663	0.973006	1.000000	0.827912
Customer_Age	0.850357	0.849160	0.827912	1.000000



```
# 8. SALES BY CATEGORY
# =====

print("\n" + "=" * 60)
print("8. SALES BY CATEGORY")
print("=" * 60)

category_sales = df.groupby("Category")["Sales"].sum()

print("\nSales by Category:")
print(category_sales)

plt.figure(figsize=(8, 5))

sns.barplot(
    x=category_sales.index,
    y=category_sales.values
)

plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
```

```
plt.tight_layout()
plt.show()
```

8. SALES BY CATEGORY

Sales by Category:

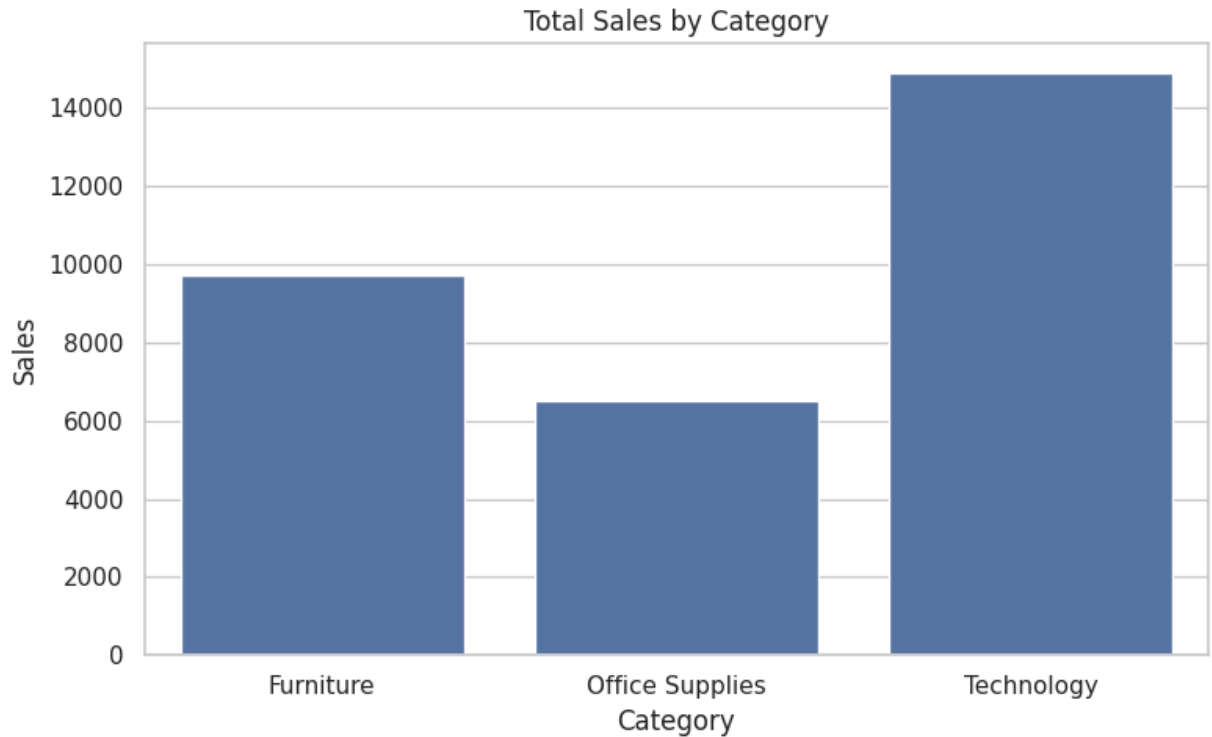
Category

Furniture 9700

Office Supplies 6500

Technology 14900

Name: Sales, dtype: int64



9. PROFIT BY REGION

=====

```
print("\n" + "=" * 60)
```

```
print("9. PROFIT BY REGION")
```

```
print("=" * 60)
```

```
region_profit = df.groupby("Region")["Profit"].sum()
```

```
print("\nProfit by Region:")
```

```
print(region_profit)
```

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(
    x=region_profit.index,
    y=region_profit.values
)
```

```
plt.title("Total Profit by Region")
```

```
plt.xlabel("Region")
```

```
plt.ylabel("Profit")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
=====
9. PROFIT BY REGION
=====
```

Profit by Region:

Region

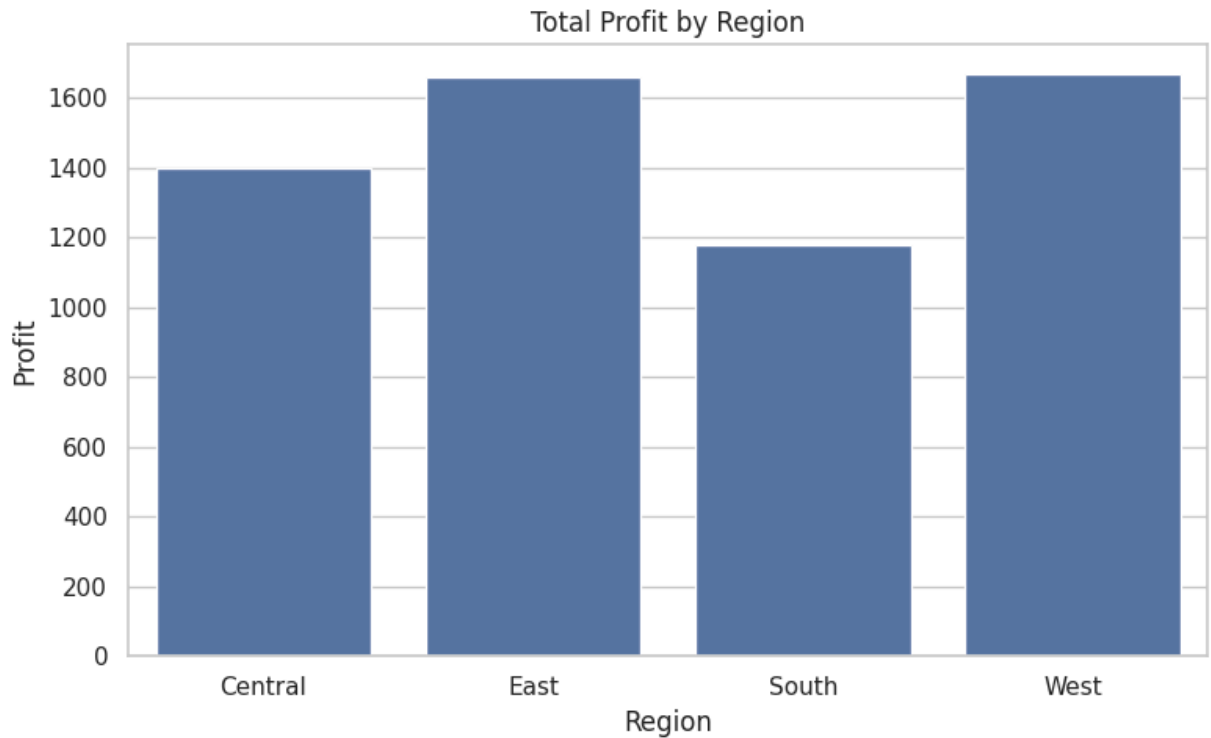
Central 1400

East 1660

South 1180

West 1670

Name: Profit, dtype: int64



```
# 10. MULTIPLE VISUALIZATIONS
```

```
# =====
```

```
print("\n" + "=" * 60)
print("10. ADDITIONAL VISUALIZATION")
print("=" * 60)
```

```
plt.figure(figsize=(8, 5))
```

```
sns.scatterplot(
    data=df,
    x="Customer_Age",
    y="Sales",
    hue="Region",
    s=100
)
```

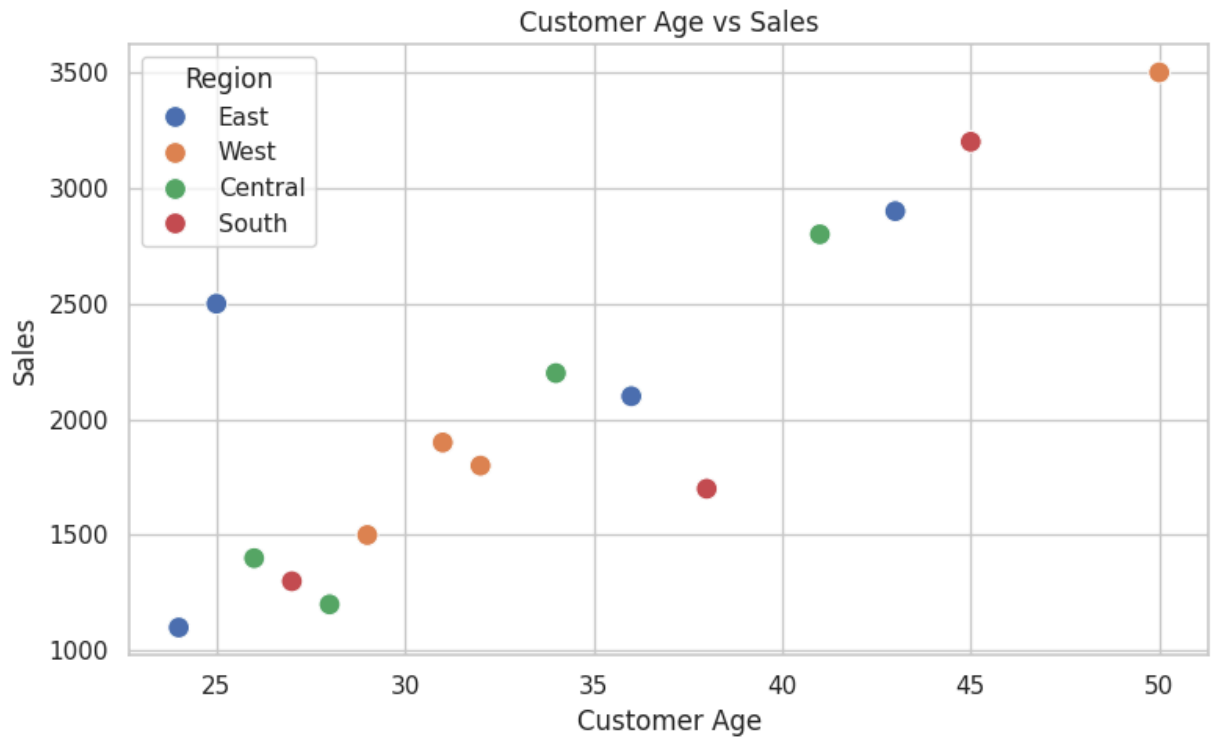
```
plt.title("Customer Age vs Sales")
plt.xlabel("Customer Age")
plt.ylabel("Sales")
plt.legend(title="Region")
```

```
plt.tight_layout()
plt.show()
```

=====

10. ADDITIONAL VISUALIZATION

=====



```
# =====
# 11. BUSINESS INSIGHTS
# =====

print("\n" + "=" * 60)
print("11. BUSINESS INSIGHTS")
print("=" * 60)

# Create required calculations first
region_sales = df.groupby("Region")["Sales"].sum()

region_profit = df.groupby("Region")["Profit"].sum()

category_sales = df.groupby("Category")["Sales"].sum()

# Find highest sales region
highest_sales_region = region_sales.idxmax()
highest_sales_value = region_sales.max()

# Find highest profit region
highest_profit_region = region_profit.idxmax()
highest_profit_value = region_profit.max()

# Find highest sales category
highest_category = category_sales.idxmax()
highest_category_value = category_sales.max()

# Display results
print("\nHighest Sales Region:")
print(highest_sales_region, "=", highest_sales_value)

print("\nHighest Profit Region:")
print(highest_profit_region, "=", highest_profit_value)
```

```
print("\nHighest Sales Category:")
print(highest_category, "=", highest_category_value)
```

```
print("""
Main Visualization Findings:
```

1. The bar chart helps compare total sales between regions.
 2. The line chart shows how sales change over different periods.
 3. The histogram shows the distribution of sales values.
 4. The box plot helps identify the spread and possible outliers.
 5. The scatter plot shows the relationship between sales and profit.
 6. The correlation heatmap shows the strength of relationships between numerical variables.
 7. Category and regional charts help identify the best-performing business areas.
 8. Data visualization makes complex data easier to understand and supports better business decision-making.
- ```
""")
```

```
=====
11. BUSINESS INSIGHTS
=====
```

```
Highest Sales Region:
West = 8700
```

```
Highest Profit Region:
West = 1670
```

```
Highest Sales Category:
Technology = 14900
```

```
Main Visualization Findings:
```

1. The bar chart helps compare total sales between regions.
2. The line chart shows how sales change over different periods.
3. The histogram shows the distribution of sales values.
4. The box plot helps identify the spread and possible outliers.
5. The scatter plot shows the relationship between sales and profit.
6. The correlation heatmap shows the strength of relationships between numerical variables.
7. Category and regional charts help identify the best-performing business areas.